

CHEMISTRY (THEORY) – MOCK PAPER SET 2

Statistically Analyzed High-Frequency Questions

Time: 3 Hours

Max Marks: 70

By Pragya

General Instructions:

- **Section A:** 16 Multiple Choice Questions (1 Mark each).
- **Section B:** 5 Very Short Answer Questions (2 Marks each).
- **Section C:** 7 Short Answer Questions (3 Marks each).
- **Section D:** 2 Case-Based Questions (4 Marks each).
- **Section E:** 3 Long Answer Questions (5 Marks each).

SECTION A (1 Mark Each)

1. Which of the following solutions will have the highest boiling point?

- (a) 1.0 M NaOH (b) 1.0 M Na₂SO₄
(c) 1.0 M NH₄NO₃ (d) 1.0 M KNO₃

Repeated Concept: Van't Hoff factor i

2. Which transition metal ion is colourless?

- (a) Ti³⁺ (b) Sc³⁺
(c) Mn²⁺ (d) Fe³⁺

Repeated Concept: d-d transition/unpaired electrons

3. The reaction $\text{Ar}-\text{N}_2^+ \text{X}^- \longrightarrow \text{Ar}-\text{CN} + \text{N}_2$ (Reagent: CuCN/KCN) is known as:

- (a) Sandmeyer's reaction (b) Gattermann reaction
(c) Friedel-Crafts reaction (d) Stephen reaction

Repeated in 56/5/1 & HFG1E

4. For a zero-order reaction, the slope of the plot of concentration of reactant vs time is:

- (a) $-k$ (b) k
(c) $k/2.303$ (d) $-k/2.303$

Repeated Graph Concept

5. Which of the following cell was used in the Apollo space programme?

- (a) Mercury cell (b) Daniel cell
(c) H₂-O₂ Fuel cell (d) Dry cell

Repeated in 56/5/1 & HFG1E

6. The specific sequence in which amino acids are arranged in a protein is called its:

- (a) Primary structure
- (b) Secondary structure
- (c) Tertiary structure
- (d) Quaternary structure

7. Identify the chiral molecule among the following:

- (a) 2-Bromobutane
- (b) 1-Bromobutane
- (c) 2-Bromopropane
- (d) 2-Bromo-2-methylpropane

8. Reaction of Glucose with HI gives:

- (a) Gluconic acid
- (b) Saccharic acid
- (c) n-Hexane
- (d) Acetyl chloride

9. Racemisation occurs in:

- (a) S_N2 reaction
- (b) S_N1 reaction
- (c) Neither S_N1 nor S_N2
- (d) Both S_N1 and S_N2

10. Transition metals act as catalysts because:

- (a) They form alloys
- (b) They show variable oxidation states
- (c) High melting points
- (d) They are malleable

11. **Assertion (A):** p-nitrophenol is more acidic than phenol.

Reason (R): Nitro group is electron withdrawing (-R effect) which stabilizes the phenoxide ion.

- (a) Both A and R are true, R explains A
- (b) Both A and R are true, R doesn't explain A

12. **Assertion (A):** units of rate constant for a zero order reaction are $\text{mol L}^{-1} \text{s}^{-1}$.

Reason (R): For zero order, Rate = k.

- (a) Both A and R are true, R explains A

13. Which alkyl halide reacts most rapidly by S_N1 mechanism?

- (a) CH_3Cl
- (b) $(\text{CH}_3)_2\text{CHCl}$
- (c) $(\text{CH}_3)_3\text{CCl}$
- (d) $\text{CH}_3\text{CH}_2\text{Cl}$

14. The magnetic moment of $[\text{NiCl}_4]^{2-}$ is approximately:

- (a) 0 BM
- (b) 2.82 BM
- (c) 1.73 BM
- (d) 5.9 BM

15. Which reagent is used to distinguish between Primary, Secondary, and Tertiary amines?

- (a) Tollen's Reagent
- (b) Hinsberg's Reagent
- (c) Fehling's Solution
- (d) Lucas Reagent

16. Colligative properties depend on:

- (a) Nature of particles
- (b) Number of solute particles
- (c) Physical properties
- (d) Nature of solvent

SECTION B (2 Marks Each)

17. (a) State **Kohlrausch's law** of independent migration of ions.
(b) How does the conductivity of a solution vary with dilution?

Source: Repeated in 56-5-1, 56-7-1, HFG1E

18. Explain the following terms with an example:

- (a) Ambidentate ligand.
(b) Linkage Isomerism.

19. **Biomolecules:** Write the reactions of D-Glucose with:

- (i) Bromine water ($\text{Br}_2/\text{H}_2\text{O}$)
(ii) Hydrogen Cyanide (HCN)

20. **Give reasons:**

- (i) Zn, Cd and Hg are generally not considered as transition metals.
(ii) Transition metals form interstitial compounds.

21. **Solutions:** What is meant by **positive deviation** from Raoult's law? Give one example of a solution showing positive deviation.

SECTION C (3 Marks Each)

22. **Chemical Kinetics (Numerical):**

Show that for a first-order reaction, the time required for **99% completion is twice** the time required for the completion of **90%** of the reaction.

23. **Electrochemistry (Numerical):**

Calculate the emf (E_{cell}) of the following cell at 298 K:



Given: $E^\circ_{\text{Mg}^{2+}/\text{Mg}} = -2.37 \text{ V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$.

24. **Coordination Compounds:** Using Valence Bond Theory (VBT), explain geometry and magnetic behavior of:

- (i) $[\text{Ni}(\text{CN})_4]^{2-}$ (At. No. Ni = 28)
(ii) $[\text{NiCl}_4]^{2-}$

Why is one diamagnetic and the other paramagnetic?

25. **Mechanism & Reactivity:**

- (i) Which alkyl halide in the pair ($\text{CH}_3\text{CH}_2\text{Br}$ or $(\text{CH}_3)_3\text{C}-\text{Br}$) reacts faster by $\text{S}_{\text{N}}1$ mechanism and why?
(ii) What happens to the optical activity (stereochemistry) during an $\text{S}_{\text{N}}1$ reaction?

26. Solutions (Numerical):

Calculate the mass of Ascorbic acid ($\text{C}_6\text{H}_8\text{O}_6$) to be dissolved in 75 g of acetic acid to lower its melting point by 1.5°C . ($K_f = 3.9 \text{ K kg mol}^{-1}$).

27. Name Reactions: Write the chemical equations for:

- (i) **Cannizzaro reaction**
- (ii) **Hoffmann Bromamide degradation**
- (iii) **Sandmeyer's reaction**

28. d-Block Trends: Account for the following:

- (i) $E^\circ (\text{Mn}^{3+}/\text{Mn}^{2+})$ is highly positive (+1.57 V) compared to $\text{Cr}^{3+}/\text{Cr}^{2+}$.
- (ii) Enthalpies of atomization of transition metals are high.
- (iii) Transition metals form coloured ions.

SECTION D (Case-Based – 4 Marks Each)

29. Case Study: Crystal Field Theory (CFT)

The colour of coordination compounds arises from the excitation of electrons from lower energy d-orbitals to higher energy d-orbitals (d–d transition). The energy difference depends on the ligand strength.

- (a) Explain why $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is violet but becomes colourless upon heating.
- (b) What is the relationship between wavelength of light absorbed and splitting energy (Δ_o)?
- (c) Arrange the ligands H_2O , NH_3 , CN^- in increasing order of field strength.

30. Case Study: Nucleophilic Substitution

Nucleophilic substitution in haloalkanes proceeds via $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ mechanisms. The choice depends on the structure of the alkyl halide and the solvent polarity. Polar protic solvents favor $\text{S}_{\text{N}}1$.

- (a) Why does $\text{S}_{\text{N}}1$ reaction rate increase with the polarity of the solvent?
- (b) Why is allylic halide highly reactive towards $\text{S}_{\text{N}}1$ reaction?
- (c) Which mechanism results in the inversion of configuration?

SECTION E (5 Marks Each)

31. (a) Electrochemistry & Kinetics:

- (i) Conductivity of 0.001 M acetic acid is $4 \times 10^{-5} \text{ S/cm}$. Calculate the degree of dissociation (a) if Λ_m° is $390 \text{ S cm}^2 \text{ mol}^{-1}$.
- (ii) Define order of reaction. Plot a graph of conc vs time for zero-order.

OR

(b) d- and f-Block Elements:

- (i) Explain the cause and consequences of **Lanthanoid Contraction**.
- (ii) Complete the reaction: $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow ?$
- (iii) Why is HCl not used to acidify KMnO_4 solution?

32. (a) Organic Compounds containing Oxygen:

- (i) An organic compound 'A' ($C_5H_{10}O$) forms a 2,4-DNP derivative, reduces Tollen's reagent, and undergoes Cannizzaro reaction. Identify 'A'.
- (ii) Write structure of product formed when 'A' reacts with Dilute NaOH.
- (iii) Distinguish Propanal and Propanone using a chemical test.

OR

(b) Distinction & Acidity:

- (i) Arrange in increasing acidity: Ethanol, Phenol, Benzoic acid, Chloroacetic acid.
- (ii) Give a test to distinguish: Phenol and Ethanol; Benzoic acid and Ethyl benzoate.
- (iii) Write the mechanism of hydration of ethene to yield ethanol.

33. Organic Compounds containing Nitrogen (Amines):

- (i) Write the reaction for **Carbylamine test**.
- (ii) Arrange in increasing basic strength in **aqueous solution**:
 CH_3NH_2 , $(CH_3)_2NH$, $(CH_3)_3N$.
- (iii) How will you convert: Nitrobenzene to Aniline AND Aniline to Phenol?

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